

Preliminary Topic Proposal

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Topic of Study: To what extent do MLB-level sabermetric strategies apply to high school baseball?

I. Introduction and Review of Literature

This research asks: to what extent do MLB-level sabermetric strategies and metrics apply to high school baseball? Sabermetrics has reshaped how professional organizations evaluate talent and manage workload, and that influence is trickling into amateur baseball even though MLB metrics are built from MLB conditions, professional athletes, 162-game seasons, and tracking infrastructure a high school season cannot replicate. The literature suggests the fit is only partial: Fleisig et al. (1999) found pitching mechanics stay consistent across levels while velocity output scales sharply with level; Scarborough et al. (2021) found an injury-risk pattern appears in high schoolers almost as often as in professionals; Mair et al. (2022), Matsuura et al. (2021), and Kriz et al. (2026) each found blanket MLB-style workload rules only work once broken into level-adjusted components; Caporale and Collier (2013) and McCue et al. (2019) show high school statistics alone poorly predict MLB readiness, which Efron and Morris (1977) explain statistically as a small-sample problem; and Thomay (2014) shows sabermetric formulas transfer across levels only with level-specific input data. No source combines these findings into one usable framework, that gap is what this research fills. My own team went second- and third-best in team ERA in our district last season and still missed the playoffs, a gap between numbers and outcome I could not let go.

II. Problem Statement & Target Demographic

The problem: high school programs are adopting MLB-derived benchmarks faster than anyone verifies whether those numbers hold at their level, and the programs least able to verify that gap are the ones adopting it uncritically, a coach citing an MLB pitch-count threshold has no way to confirm it was ever validated on a teenage arm, since the tracking infrastructure that generates those numbers (a TrackMan unit runs roughly \$30,000; a FlightScope Strike, \$18,000) is out of reach for a typical public program (Driveline Baseball, 2016; The Hardball Times, 2017). The target demographic is under-resourced U.S. high school baseball programs, public teams without booster-funded access to even consumer tools like Rapsodo or HitTrax, whose cost and required subscription still price out most athletic budgets (Rapsodo, n.d.), a widening “data divide” between well-funded travel programs and public high schools (Business of Ball, 2025; Project Play, 2025). The impact: coaches make starting, recruiting, and workload decisions on unvalidated MLB intuition (Mair et al., 2022; Matsuura et al., 2021), and players on programs like mine can be genuinely strong and still get misread, because the numbers were never built for us.

III. Researcher Qualifications

I bring both statistical training and direct baseball experience to this research. I've completed AP Statistics, AP Calculus AB/BC, and Calculus III, along with Sabermetrics 101 and a SABR Sabermetrics Certification, giving me the regression and hypothesis-testing background needed to weight competing success factors rather than just describe them. I use Python for statistical modeling on two active projects, a high school umpire strike-call tendency model and a physics-based model of exit velocity, launch angle, and trajectory, neither of which depends on Rapsodo- or TrackMan-level hardware. As a current varsity pitcher, I also bring firsthand knowledge of what high school competition and coaching decisions actually look like. Together, these let me build a statistically weighted, high-school-specific model on accessible inputs, test which MLB metrics actually hold predictive value at this level, and keep it grounded in what a real coach could use.

IV. Conclusion

MLB analytics has transformed how baseball is understood, but that revolution has outpaced the verification work needed to responsibly bring it down to high school baseball, leaving under-resourced programs adopting numbers never built or tested for them. This research does not dismiss analytics, the literature shows real value in physics-based modeling and workload design when correctly adjusted, but argues that adjustment must be done deliberately, factor by factor, not assumed. My own team's playoff miss despite an elite team ERA is a small example of that larger gap. If this research succeeds, it gives coaches and players at programs like mine, without a six-figure tracking budget, a genuinely usable way to know which numbers to trust, adjust, or set aside.

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